Quantification: 536-63x_Hyp_RD_z08c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	27
interior lacuna count	24
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 24)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2617	2458.5	1727.14	px^2
major_axis_length_px	lacuna shape	92.4973	99.3629	42.6278	px
minor_axis_length_px	lacuna shape	39.7853	38.7847	14.5046	px
aspect_ratio	lacuna shape	2.3245	1.9781	0.8851	unitless
eccentricity	lacuna shape	0.8543	0.8626	0.1031	unitless
solidity	lacuna shape	0.7729	0.7971	0.1213	unitless
roots_count	headline	9.5	9	5.801	count
ring_length_r30_px	headline	282.625	273.5	165.599	px
ring_length_r60_px	reported	555.417	539	334.914	px
owned_length_px	ownership-dependent	715.901	597.337	574.541	px
edge_count	ownership-dependent	35.0417	31	28.4276	count
mean_edge_length_px	ownership-dependent	21.4821	20.8255	6.5554	px
perimeter_px	normalised	300.902	267.146	152.89	px
ring_area_r30_px2	normalised	8533.96	9371.5	3553.71	px^2
ring_area_r60_px2	normalised	18223.7	18857	8375.7	px^2
in_frame_fraction_r30	normalised	0.9882	1	0.0401	unitless
in_frame_fraction_r60	normalised	0.971	1	0.0784	unitless
ring_density_r30	normalised	0.0316839	0.0310934	0.00889289	px^-1
ring_density_r60	normalised	0.029505	0.0289221	0.00730896	px^-1
roots_per_100px_perimeter	normalised	3.1693	3.1926	1.1488	per 100 px
ring_attached_length_r30_px	network v2	253.375	245.5	157.25	px
ring_attached_length_r60_px	network v2	428.458	415.5	293.111	px
ring_length_w_r30_px	network v2	319.951	309.836	185.163	px
ring_length_w_r60_px	network v2	632.568	613.272	375.396	px
sholl_crossings_r10	network v2	7.4583	8	3.3098	count
sholl_crossings_r20	network v2	7.3333	7	4.2902	count
sholl_crossings_r30	network v2	7.5	6	4.1911	count
loop_count	loops, report only	1.75	1	2.5065	count
loop_edge_share	loops, report only	0.1347	0.1235	0.1241	share
ring_width_mean_r30_px	width	7.5919	7.3605	0.8529	px

Per field
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measure	value	units
analysed_area_px2	945710	px^2
total_skeleton_length_px	25398	px
canalicular_length_density_per_px	0.026856	px^-1
junction_count	491	count
junction_density_per_px2	0.00051919	px^-2
lacuna_count	27	count

lacunae_per_px2	2.855e-05	px^-2
skeleton_component_count	355	count
mean_component_length_px	71.5437	px
median_component_length_px	24	px
field_length_density_w_per_px	0.0305105	px^-1
field_density_without_flagged_per_px	0.0268991	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	270	px
unattached_component_count	16	count
canalicular_length_density_attached_per_px	0.026856	px^-1
field_length_density_with_unattached_per_px	0.0271415	px^-1
loop_count	53	count
loop_edge_share	0.1532	share
edge_count	1325	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8266	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	25102	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	468	33.3861	22.2869	1.498	0.7446	0.7647	0.8461	10.1432	885.363	True
2	2181	148.407	31.7971	4.6673	0.9768	0.5491	-0.2241	127.792	674.16	False
3	3003	146.448	39.2919	3.7272	0.9633	0.5868	-0.1342	219.3	634.066	False
4	4348	102.17	64.0204	1.5959	0.7793	0.7299	-0.8016	123.509	117.145	False
5	6103	130.52	74.6261	1.749	0.8204	0.7441	-0.6975	211.602	115.526	False
6	4713	113.091	66.6058	1.6979	0.8082	0.6869	0.0149	166.949	811.934	False
7	2838	120.116	56.4992	2.126	0.8825	0.5006	-0.0922	183.402	482.588	False
8	640	32.9351	28.0278	1.1751	0.5252	0.8112	-0.2136	241.733	39.7797	False
9	593	40.0782	21.7723	1.8408	0.8396	0.7583	0.4372	249.45	834.877	False
10	5444	169.703	43.0843	3.9389	0.9672	0.8492	-0.1663	315.438	989.012	False
11	472	35.2068	21.2692	1.6553	0.7969	0.7528	-0.5629	318.485	581.102	False
12	5032	145.049	50.087	2.8959	0.9385	0.6975	-0.3567	389.803	259.63	False
13	1325	56.519	31.2238	1.8101	0.8335	0.9138	-0.4181	431.483	840.339	False
14	1869	63.382	43.4291	1.4594	0.7284	0.783	-0.4231	489.012	823.251	False
15	4023	130.997	44.5634	2.9396	0.9404	0.8529	-0.2947	478.273	373.626	False
16	2063	91.1753	31.0575	2.9357	0.9402	0.8892	-0.3885	511.078	505.802	False
17	38997	474.656	126.173	3.7619	0.964	0.8704	-0.1612	856.424	971.407	True
18	2736	102.242	36.178	2.8261	0.9353	0.8653	-0.4465	632.235	931.07	False
19	2738	96.5558	37.4594	2.5776	0.9217	0.8784	-0.3793	679.917	399.446	False
20	3288	105.667	42.059	2.5123	0.9174	0.8761	-0.3815	724.683	177.336	False
21	797	46.6745	23.1748	2.014	0.868	0.8935	-0.2375	760.548	828.605	False
22	584	33.791	23.1054	1.4625	0.7297	0.9125	-0.7852	793.788	813.11	False
23	1183	68.5981	39.5227	1.7357	0.8173	0.5796	-0.4587	845.742	872.086	False
24	518	37.316	23.4561	1.5909	0.7777	0.7443	0.7533	887.49	867.025	False

25	4334	128.945	44.2583	2.9135	0.9393	0.8796	-0.1396	903.815	374.562	False
26	1983	74.3475	38.2775	1.9423	0.8573	0.8154	-0.0273	970.974	775.836	False
27	593	45.6756	20.4671	2.2317	0.894	0.7907	1.5552	1015.45	766.083	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	105	215	202.894	9	22.5438	7.3737
2	21	392	645	542.245	38	14.2696	7.9442
3	14	291	498	731.097	40	18.2774	7.1502
4	8	341	587	577.387	31	18.6254	7.2773
5	10	261	583	617.286	23	26.8385	7.566
6	20	457	783	859.595	40	21.4899	7.528
7	19	723	1416	1324.53	81	16.3522	6.6187
8	5	148	269	487.375	25	19.495	6.7585
9	6	202	455	726.612	31	23.4391	7.3169
10	10	325	580	1119.45	52	21.5278	7.3803
11	5	236	415	172.51	11	15.6827	6.8321
12	20	632	1263	2681.43	133	20.1611	7.0906
13	4	90	197	79.1838	7	11.312	7.454
14	8	165	372	227.338	12	18.9448	6.9494
15	13	411	890	1092.82	61	17.9151	8.8041
16	10	286	601	967.637	42	23.039	8.2735
17	5	302	559	236.581	9	26.2867	7.0501
18	6	228	401	566.186	25	22.6474	9.0318
19	11	305	670	1297.54	48	27.0322	8.1069
20	10	349	860	1189.62	50	23.7924	7.3406
21	4	91	213	230.208	16	14.388	6.8986
22	2	94	143	75.8406	2	37.9203	8.0227
23	7	187	245	405.961	26	15.6139	6.6557
24	1	44	91	114.983	3	38.3276	9.9543
25	10	375	855	842.269	33	25.5233	8.5809
26	4	150	298	252.522	11	22.9565	6.6699
27	2	13	13	13.6569	1	13.6569	3.8495

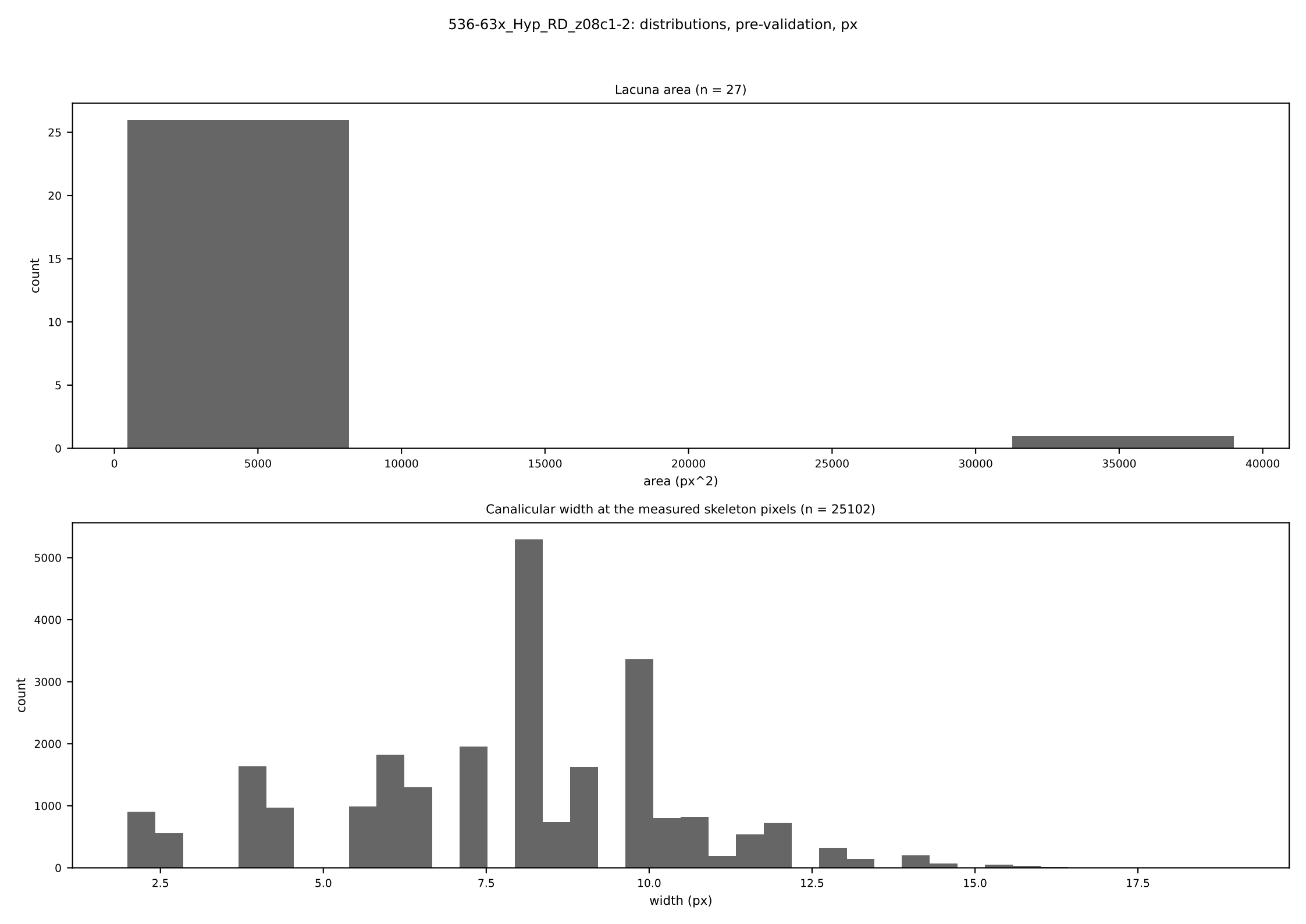
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	111.598	3553	9454	0.6134	0.5737	0.0295525	0.0227417	5.3764
2	415.641	10082	20733	1	1	0.0388812	0.0311098	5.0524
3	495.973	9778	17737	1	1	0.0297607	0.0280769	2.8227
4	488.535	9556	20455	1	1	0.0356844	0.0286971	1.6375
5	609.269	10724	19977	1	0.9991	0.0243379	0.0291836	1.6413
6	442.446	12800	26204	1	1	0.0357031	0.0298809	4.5203
7	514.931	14099	31059	1	1	0.0512802	0.0455907	3.6898
8	123.439	3719	6442	0.9752	0.821	0.0397956	0.0417572	4.0506
9	128.539	4684	10910	1	1	0.0431255	0.0417049	4.6678
10	423.747	11743	23622	0.8152	0.7003	0.0276761	0.0245534	2.3599
11	117.539	5258	12354	1	1	0.044884	0.0335924	4.2539
12	487.973	14467	32198	1	1	0.0436856	0.039226	4.0986
13	156.16	5263	12719	1	1	0.0171005	0.0154886	2.5615
14	228.202	6707	15935	1	1	0.0246012	0.0233448	3.5057
15	348.433	12410	27080	1	1	0.0331185	0.0328656	3.731
16	227.38	9187	22623	1	1	0.031131	0.0265659	4.3979
17	1282.44	13782	18371	0.4486	0.431	0.0219126	0.0304284	0.3899
18	274.735	8709	16073	1	1	0.0261798	0.0249487	2.1839
19	250.693	9821	24082	1	1	0.0310559	0.0278216	4.3878

20	298.534	10681	26849	1	1	0.0326748	0.032031	3.3497
21	131.776	4471	9529	1	1	0.0203534	0.0223528	3.0355
22	102.847	3623	7408	1	1	0.0259453	0.0193035	1.9446
23	259.557	5651	7254	1	1	0.0330915	0.0337745	2.6969
24	120.119	2651	3753	1	1	0.0165975	0.0242473	0.8325
25	340.877	12243	29334	1	0.982	0.0306298	0.0291471	2.9336
26	234.309	6488	13038	0.9258	0.8017	0.0231196	0.0228563	1.7071
27	129.012	1451	2663	0.5828	0.5538	0.00895934	0.00488171	1.5502

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	99	171	122.225	246.865	2	3	3
2	342	438	441.747	736.24	8	12	10
3	279	358	330.007	564.245	8	6	5
4	273	470	389.676	672.541	10	11	9
5	248	452	292.936	662.299	11	7	6
6	428	712	517.546	894.495	10	10	12
7	694	1052	813.127	1593.25	14	11	16
8	92	116	170.752	314.12	6	5	4
9	202	435	228.924	526.002	6	3	5
10	325	552	368.291	663.198	9	10	7
11	179	179	272.35	480.002	6	3	5
12	563	1136	709.985	1430.87	13	16	16
13	76	76	102.326	221.995	4	0	2
14	150	205	184.196	422.818	3	5	5
15	371	812	458.635	1009.29	8	11	13
16	243	396	326.735	682.541	12	7	10
17	219	301	335.735	632.5	8	7	12
18	228	336	269.108	467.103	7	8	6
19	305	602	350.978	779.181	8	9	8
20	261	536	373.551	951.796	8	11	11
21	91	195	104.083	248.622	3	3	2
22	46	57	108.397	168.409	3	4	6
23	187	244	221.137	288.907	6	4	3
24	40	58	48.799	100.184	2	1	2
25	329	642	423.019	955.595	10	15	12
26	129	224	172.51	347.919	4	4	5
27	13	13	13.6569	13.6569	0	0	0


536-63x_Hyp_RD_z08c1-2: distributions, pre-validation, px
Lacuna area (n = 27)
count
area (px^2)
0
5
10
15
20
25
0
5000
10000
15000
20000
25000
30000
35000
40000
Canalicular width at the measured skeleton pixels (n = 25102)
count
width (px)
0
1000
2000
3000
4000
5000
2.5
5.0
7.5
10.0
12.5
15.0
17.5